

LEARN PYTHON PROGRAMMING – BEGINNER TO MASTER

Section: 6. Loops - Control Flow

Lecture: 60. Challenge - Factorial of Given Number

Section 1: Lecture Summary

This lecture focuses on solving two classic programming challenges using the `for` loop in Python:

1. Calculating the sum of the first n natural numbers.
2. Calculating the factorial of a number n.

The instructor revisits the use of the `range()` function with `for` loops and explains how to iterate over natural numbers properly, demonstrating the difference between summing and multiplying series of numbers. The lecture includes a detailed walkthrough of the logic, program tracing, and variable updates during the loop iterations. The lecturer also explains the importance of initializing accumulators (`sum` for addition and `fact` for multiplication) and clarifies how to handle input and output in a Python development environment like VS Code. Finally, the lecture encourages students to attempt these two challenges on their own before proceeding.

Section 2: Key Concepts and Explanations

For Loop with range() Function

- The `for` loop executes a block of code repeatedly.
- `range(start, stop)` generates numbers starting from `start` up to but not including `stop`.

- For iterating from 1 to n, use `range(1, n+1)` to include n.

Sum of First n Natural Numbers

- Natural numbers start from 1 and go upwards (1, 2, 3, ..., n).
- Problem: Calculate $sum = 1 + 2 + 3 + \dots + n$.
- Known formula: $\frac{n \times (n+1)}{2}$.
- Using a loop, initialize `sum = 0`.
- Iterate each number `i` from 1 to n and add to `sum`.

Factorial of a Number

- Definition: $n! = 1 \times 2 \times 3 \times \dots \times n$.
- Factorial is the product of all natural numbers up to n.
- Special initialization: `fact = 1` (multiplicative identity).
- Loop from 1 to n and multiply each `i` to `fact`.
- Factorial of 5 is $1 \times 2 \times 3 \times 4 \times 5 = 120$.

Variables and Iteration Tracking

- Variables `n`, `sum`/`fact`, and `i` control the computation.
- `i` iterates over each natural number.
- `sum` aggregates addition; `fact` aggregates multiplication.
- Trace tables help visualize variable updates per loop iteration.

Input and Output in Development Environments

- Use `input()` to take user input for `n`.
- If running in VS Code, use "Run Python File in Terminal" for input prompts to work properly.

- Avoid "Run Code" button in VS Code for interactive inputs, as it doesn't support terminal input.

Section 3: Example Code and Use Cases

Sum of First n Natural Numbers Using for Loop

Input: number n

```
n = int(input("Enter a number: "))
sum = 0 # Initialize sum

for i in range(1, n + 1):
    sum += i # Add i to sum

print(f"Sum of first {n} natural numbers is: {sum}")
```

This code asks the user for input and calculates the sum of natural numbers from 1 to n by iteratively adding each number.

Factorial of a Number Using for Loop

Input: number n

```
n = int(input("Enter a number to find factorial: "))
fact = 1 # Initialize factorial accumulator

for i in range(1, n + 1):
    fact *= i # Multiply fact by i

print(f"Factorial of {n} is: {fact}")
```

This code calculates the factorial of n by multiplying each number from 1 to n.

Section 4: Key Takeaways

- A `for` loop with `range(1, n+1)` is ideal to iterate through numbers from 1 to n inclusive.
- To calculate the sum of natural numbers using a loop, initialize a sum variable to 0 and repeatedly add each number.
- To calculate factorials, initialize a product variable to 1 and multiply by each number in iteration.
- Factorial grows rapidly and is the product, not sum, of natural numbers.
- Proper initialization of accumulator variables (`sum = 0`, `fact = 1`) is crucial.
- Development environments like VS Code require running the Python script in the terminal to accept input from `input()`.
- Tracing variable changes during loop iterations helps understand loop execution and debugging.

Lecture in Slides

Please Note: This document presents a curated selection of slides from the lecture; others have been excluded for conciseness.



Challenges : for Loop

Challenges : for Loop

- Sum of n Natural Numbers

Challenges : for Loop

- Sum of n Natural Numbers
- Factorial of a Number

Sum of n Natural Numbers Challenge

Sum of n Natural Numbers

Challenge

1 2 3 4 5 n

Sum of n Natural Numbers

Challenge

1 + 2 + 3 + 4 + 5 + + n

Sum of n Natural Numbers
Challenge

$$\text{Sum} = 1 + 2 + 3 + 4 + 5 + \dots + n$$

Sum of n Natural Numbers
Challenge

$$\text{Sum} = 1 + 2 + 3 + 4 + 5 + \dots + n$$

$$\text{Sum} = \frac{n * (n + 1)}{2}$$

Sum of n Natural Numbers

Challenge

$$\text{Sum} = 1 + 2 + 3 + 4 + 5 + \dots + n$$

Sum of n Natural Numbers

Challenge

$$\text{Sum} = 1 + 2 + 3 + 4 + 5 + \dots + n$$

$$n = 5$$

Sum of n Natural Numbers

Challenge

$$\text{Sum} = 1 + 2 + 3 + 4 + 5 + \dots + n$$

$$n = 5$$

$$\text{sum} = 0$$

Sum of n Natural Numbers

Challenge

$$\text{Sum} = 1 + 2 + 3 + 4 + 5 + \dots + n$$

n = 5

sum = 0

for i in range() :

Sum of n Natural Numbers

Challenge

$$\text{Sum} = 1 + 2 + 3 + 4 + 5 + \dots + n$$

n = 5

sum = 0

for i in range(1) :

Sum of n Natural Numbers

Challenge

$$\text{Sum} = 1 + 2 + 3 + 4 + 5 + \dots + n$$

n = 5

sum = 0

for i in range(1, n + 1):

Sum of n Natural Numbers

Challenge

$$\text{Sum} = 1 + 2 + 3 + 4 + 5 + \dots + n$$

n = 5

sum = 0

for i in range(1, n + 1):

sum = sum + i

Sum of n Natural Numbers

Challenge

$$\text{Sum} = 1 + 2 + 3 + 4 + 5 + \dots + n$$

```
n = 5
sum = 0
for i in range(1, n + 1):
    sum = sum + i
print('Sum of n: ', sum)
```

Sum of n Natural Numbers

Challenge

$$\text{Sum} = 1 + 2 + 3 + 4 + 5 + \dots + n$$

```
n = 5
sum = 0
for i in range(1, n + 1):
    sum = sum + i
print('Sum of n: ', sum)
```



Sum of n Natural Numbers

Challenge

Sum = 1 + 2 + 3 + 4 + 5 + + n

```
n = 5
sum = 0
for i in range(1, n + 1):
    sum = sum + i
print('Sum of n: ', sum)
```

n



Sum of n Natural Numbers

Challenge

Sum = 1 + 2 + 3 + 4 + 5 + + n

```
n = 5
sum = 0
for i in range(1, n + 1):
    sum = sum + i
print('Sum of n: ', sum)
```

n

sum



Sum of n Natural Numbers

Challenge

Sum = 1 + 2 + 3 + 4 + 5 + + n

```
n = 5
sum = 0
for i in range(1, n + 1):
    sum = sum + i
print('Sum of n: ', sum)
```

n

i

sum



Sum of n Natural Numbers

Challenge

Sum = 1 + 2 + 3 + 4 + 5 + + n

```
n = 5
sum = 0
for i in range(1, n + 1):
    sum = sum + i
print('Sum of n: ', sum)
```

n

i

sum



Sum of n Natural Numbers

Challenge

Sum = 1 + 2 + 3 + 4 + 5 + + n

n = 5

sum = 0

for i in range(1, n + 1):

sum = sum + i

print('Sum of n: ', sum)

n

5

i

sum



Sum of n Natural Numbers

Challenge

Sum = 1 + 2 + 3 + 4 + 5 + + n

```
n = 5
sum = 0
for i in range(1, n + 1):
    sum = sum + i
print('Sum of n: ', sum)
```

n

5

i

sum

--	0



Sum of n Natural Numbers

Challenge

Sum = 1 + 2 + 3 + 4 + 5 + + n

```
n = 5
sum = 0
for i in range(1, n + 1):
    sum = sum + i
print('Sum of n: ', sum)
```

n

5

i

sum

--	0
1	



Sum of n Natural Numbers

Challenge

Sum = 1 + 2 + 3 + 4 + 5 + + n

```
n = 5
sum = 0
for i in range(1, n + 1):
    sum = sum + i
print('Sum of n: ', sum)
```

n

5

i	sum
--	0
1	1
2	3
3	6
4	10
5	15



Sum of n Natural Numbers

Challenge

Sum = 1 + 2 + 3 + 4 + 5 + + n

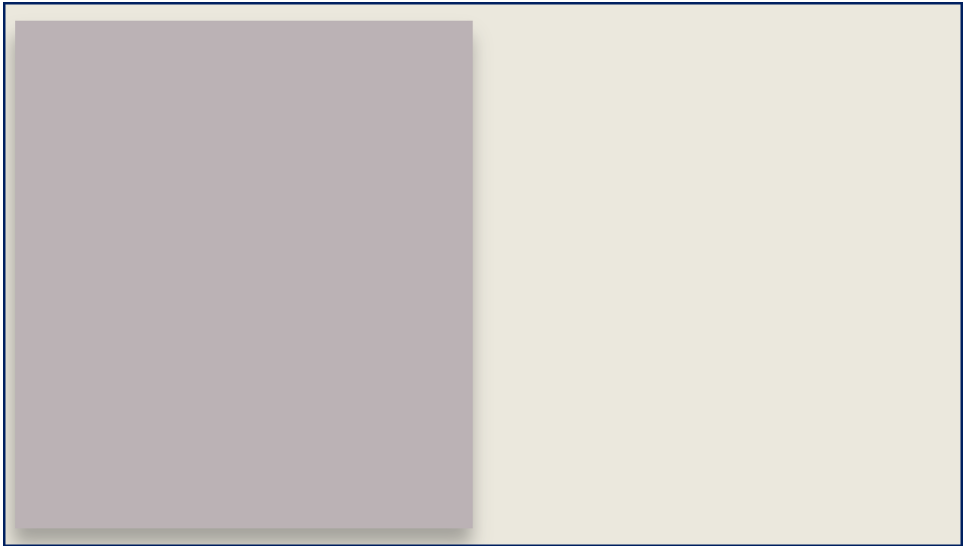
```
n = 5
sum = 0
for i in range(1, n + 1):
    sum = sum + i
print('Sum of n: ', sum)
```

n

5

i	sum
--	0
1	1
2	3
3	6
4	10
5	15





Factorial of a Numbers

Challenge

Factorial of a Numbers

Challenge

$n!$

Factorial of a Numbers

Challenge

$$n! = n$$

Factorial of a Numbers

Challenge

$$n! = n * (n-1)$$

Factorial of a Numbers

Challenge

$$n! = n * (n-1) * (n-2)$$

Factorial of a Numbers

Challenge

$$n! = n * (n-1) * (n-2) * \dots\dots\dots$$

Factorial of a Numbers

Challenge

$$n! = n * (n-1) * (n-2) * \dots\dots\dots * 2$$

Factorial of a Numbers

Challenge

$$n! = n * (n-1) * (n-2) * * 2 * 1$$

Factorial of a Numbers

Challenge

$$n! = n * (n-1) * (n-2) * * 2 * 1$$

n!

Factorial of a Numbers

Challenge

$$n! = n * (n-1) * (n-2) * * 2 * 1$$

$$n! = 1$$

Factorial of a Numbers

Challenge

$$n! = n * (n-1) * (n-2) * * 2 * 1$$

$$n! = 1 * 2$$

Factorial of a Numbers

Challenge

$$n! = n * (n-1) * (n-2) * * 2 * 1$$

$$n! = 1 * 2 * 3$$

Factorial of a Numbers

Challenge

$$n! = n * (n-1) * (n-2) * * 2 * 1$$

$$n! = 1 * 2 * 3$$

Factorial of a Numbers

Challenge

$$n! = n * (n-1) * (n-2) * \dots * 2 * 1$$

$$n! = 1 * 2 * 3 \dots * (n-1)$$

Factorial of a Numbers

Challenge

$$n! = n * (n-1) * (n-2) * \dots * 2 * 1$$

$$n! = 1 * 2 * 3 \dots * (n-1) * n$$

Factorial of a Numbers

Challenge

$$n! = 1 * 2 * 3 * (n-1) * n$$

Factorial of a Numbers

Challenge

$$n! = 1 * 2 * 3 * (n-1) * n$$

Factorial of a Numbers

Challenge

$$n! = 1 * 2 * 3 * (n-1) * n$$

$$5!$$

Factorial of a Numbers

Challenge

$$n! = 1 * 2 * 3 * (n-1) * n$$

$$5! = 1 * 2 * 3 * 4 * 5$$

Factorial of a Numbers

Challenge

$$n! = 1 * 2 * 3 * (n-1) * n$$

$$5! = 1 * 2 * 3 * 4 * 5 = 120$$

Factorial of a Numbers

Challenge

$$n! = 1 * 2 * 3 * (n-1) * n$$

$$5! = 1 * 2 * 3 * 4 * 5 = 120$$

$$\text{Sum} = 1 + 2 + 3 + 4 + 5 + + n$$

Factorial of a Numbers

Challenge

$$n! = 1 * 2 * 3 * (n-1) * n$$

$$5! = 1 * 2 * 3 * 4 * 5 = 120$$

fact

Factorial of a Numbers

Challenge

$$n! = 1 * 2 * 3 * (n-1) * n$$

$$5! = 1 * 2 * 3 * 4 * 5 = 120$$

$$\text{fact} = 1$$

Factorial of a Numbers

Challenge

$$n! = 1 * 2 * 3 * (n-1) * n$$

$$5! = 1 * 2 * 3 * 4 * 5 = 120$$

$$\text{fact} = 1 * 1$$

Factorial of a Numbers Challenge

$$n! = 1 * 2 * 3 * (n-1) * n$$

$$5! = 1 * 2 * 3 * 4 * 5 = 120$$

$$\text{fact} = 1 * 1 * 2$$

Factorial of a Numbers Challenge

$$n! = 1 * 2 * 3 * (n-1) * n$$

$$5! = 1 * 2 * 3 * 4 * 5 = 120$$

$$\text{fact} = 1 * 1 * 2 * 3$$

Factorial of a Numbers Challenge

$$n! = 1 * 2 * 3 * (n-1) * n$$

$$5! = 1 * 2 * 3 * 4 * 5 = 120$$

$$\text{fact} = 1 * 1 * 2 * 3 * 4$$

Factorial of a Numbers

Challenge

$$n! = 1 * 2 * 3 * (n-1) * n$$

$$5! = 1 * 2 * 3 * 4 * 5 = 120$$

$$\text{fact} = 1 * 1 * 2 * 3 * 4 * 5$$

Factorial of a Numbers

Challenge

$$\text{fact} = 1 * 1 * 2 * 3 * 4 * 5$$

Factorial of a Numbers

Challenge

fact = 1 * 1 * 2 * 3 * 4 * 5

n = 5

Factorial of a Numbers

Challenge

fact = 1 * 1 * 2 * 3 * 4 * 5

n = 5

fact = 1

Factorial of a Numbers

Challenge

fact = 1 * 1 * 2 * 3 * 4 * 5

n = 5

fact = 1

for i **in** range() :

Factorial of a Numbers

Challenge

fact = 1 * 1 * 2 * 3 * 4 * 5

n = 5

fact = 1

for i **in** range(1, n + 1) :

Factorial of a Numbers

Challenge

fact = 1 * 1 * 2 * 3 * 4 * 5

```
n = 5
```

```
fact = 1
```

```
for i in range(1, n + 1):
```

```
    fact = fact * i
```

Factorial of a Numbers

Challenge

fact = 1 * 1 * 2 * 3 * 4 * 5

```
n = 5
```

```
fact = 1
```

```
for i in range(1, n + 1):
```

```
    fact = fact * i
```

```
print( 'Factorial of n: ', fact )
```

